3.2 Parallel Lines and Transversals

Name _____ Period ____ Date _____

Geometry 3.2 · Student sheet

I can…

- use corresponding angles to find angle measures when lines are parallel.
- use alternate interior, alternate exterior, and consecutive interior angles to find measures.
- prove a theorem about parallel lines.

Vocabulary · fill in the missing word

Corresponding angles — angles in the same position at the two _____ of a transversal

Alternate interior angles — angles between the two lines on opposite sides of the _____

Alternate exterior angles — angles outside the two lines on opposite sides of the _____

Consecutive interior angles — angles between the two lines on the same side of the _____

Part 1 · Discovery: what is equal?

Use two parallel lines of your notebook paper and a transversal. Measure, then sort.

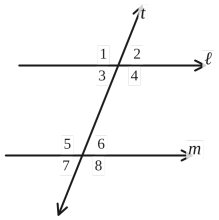
Kind of pair	Example pair	Measures	Equal or add to 180°?
Corresponding			
Alternate interior			
Alternate exterior			
Consecutive interior			

Part 2 · Notes and examples (fill in each note, then work the example)

Corresponding Angles Theorem: if two parallel lines are cut by a transversal, then corresponding angles are _____.

Example 1

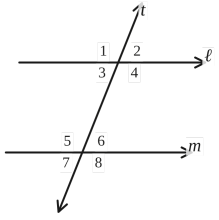
$\ell \parallel m$ and $m\angle 1 = 115°$. Find $m\angle 5$ and $m\angle 8$.



Alternate Interior and Alternate Exterior Angles Theorems: these pairs are also _____ when the lines are parallel.

Example 2

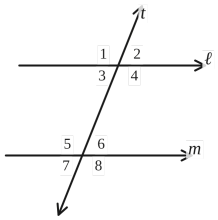
$\ell \parallel m$ and $m\angle 3 = 68°$. Find $m\angle 6$, $m\angle 2$, and $m\angle 7$.



Consecutive Interior Angles Theorem: if two parallel lines are cut by a transversal, then consecutive interior angles are _____.

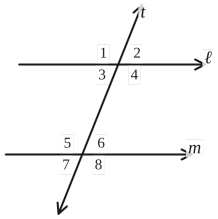
Example 3

$\ell \parallel m$, $m\angle 4 = (3x + 10)°$, and $m\angle 6 = (2x + 20)°$. Find x and both angles.



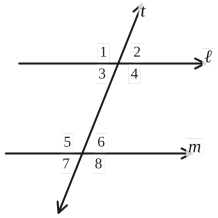
Example 4

$\ell \parallel m$, $m\angle 3 = (4x - 15)°$, and $m\angle 6 = (2x + 35)°$. Find x and the angle measure.



Example 5

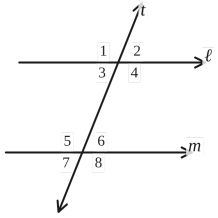
Given: $\ell \parallel m$. Prove: $\angle 3 \cong \angle 6$ (Alternate Interior Angles Theorem).



Part 3 · Partner practice

Partner 1

$\ell \parallel m$ and $m\angle 2 = 72°$. Find $m\angle 6$ and $m\angle 5$.



Partner 2

$\ell \parallel m$, $m\angle 1 = (6x - 8)°$, and $m\angle 8 = (4x + 22)°$. Find x and the angle measure.

